

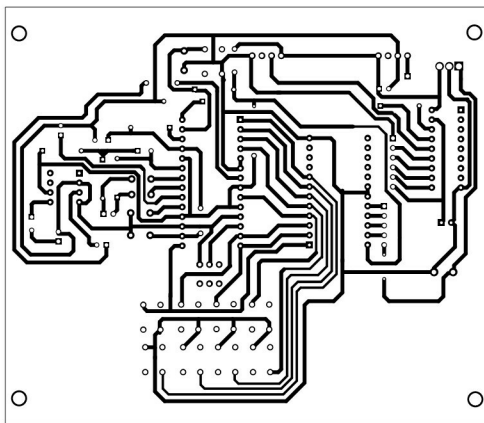


Fig. 5: Actual-size PCB pattern of the receiver section

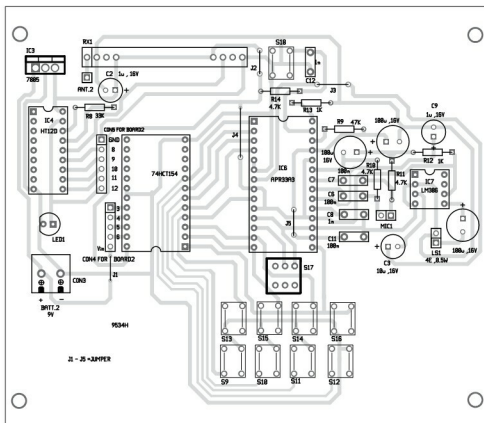


Fig. 6: Component layout of the receiver section

Test Points

Test point	Details
TP0, TP5	0V, GND
TP1, TP4	+9V DC
TP2, TP6	+5V DC
TP3	Serial data to TX1
TP7	Serial data from RX1

EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com

pressed key. The code is modified to work for 16 switches for your convenience, so that if you want to increase the message switches and keypad switches apart from those mentioned in the PCB designs of the project, you will simply have to modify the hardware, and no changes would be required in the software section.

Initially, the code configures the column pins and row pins for the keypad and then in the setup() function, it configures the output pins for Arduino UNO Board1.

Loop function is used to read the keypad and Digital Write() function is used to output the data bits corresponding to read character.

In the receiver section code, I/O pins are configured for Arduino UNO Board2 and then the serial port is enabled for viewing the pressed key info on the serial monitor as well.

Loop() function is used to read the receiving bits and convert these from binary to decimal values and send the received data to the output pins. ●

Poonguzhali S. is associate professor at Department of Electronics and Communication Engineering, College of Engineering, Anna University, Guindy, Chennai. All other authors are biomedical engineering students at College of Engineering